



Utilising Machine Learning to Optimise the Identification of Metal Organic Frameworks in Hydrogen Storage

MMME3083 INDIVIDUAL BENG PROJECT INTERIM REPORT

Author: Nina So

Student ID: 20475904

Date of report: 7th Jan 2024

Supervisor: Sanliang Ling

Summary

This report introduces a bachelor's individual project that will aim to identify potential metal organic frameworks (MOFs) for use in hydrogen storage via the use of a neural network. The neural network will identify materials with desirable properties for the storage of hydrogen. In doing so, it will reduce the factor of trial and error in the testing of potential materials for hydrogen storage as well as the cost of money and time in manufacturing the materials. This will support the move towards a more sustainable energy option in sectors such as transport for commercial businesses in cargo ships to personal vehicles.

Current progress of the project includes: a literature review on previous neural networks focusing on various regression models created to analyse the structure of MOFs; the production of code to featurise any data expected to be parsed into the network; and the initial planning of the first algorithm to be programmed. Compared to the planning documentation, the project is 2 weeks behind schedule as only the planning of the code has been started and not the development cycles of the code. The previous Gantt chart has been included in Appendix 1.

The literature review has highlights the main limitations of current studies.

The code made to featurise the data will evolve depending on the needs of the neural network and will be adapted in parallel. At present, the code has been deemed sufficient.

Contents

NOMENCLATURE.....	Error! Bookmark not defined.
ACRONYMS	3
1. BACKGROUND AND CONTEXT	4
1.1. Aim and Objectives	6
2. LITERATURE REVIEW	7
3. PROGRESS UPDATE	9
3.1. Methodology	9
3.2. Featurisation	9
3.3. Program Planning	10
3.4. Preliminary results and data analysis.....	Error! Bookmark not defined.
4. PLANNING UPDATE	10
5. CONCLUSIONS.....	13
6. REFERENCES	14
APPENDIX 3 – Other relevant Material	Error! Bookmark not defined.

ACRONYMS

Acronym	Meaning
MOF	Metal organic framework
ERT	Extremely randomised trees
PS	Pressure swing
TPS	Temperature and pressure swing

1. BACKGROUND AND CONTEXT

Global warming still currently proves to be a prominent issue in our lives. From threatening fragile ecosystems to intensifying the effects of natural disasters, its impact is witnessed by all. More countries are pushing for sustainable lifestyles to combat global warming. In doing so, the output of carbon dioxide (and other greenhouse gases that trap heat within the Earth's atmosphere) is reduced. The UK government aims to reach net-zero by 2050 by replacing "with cheaper, cleaner, domestic sources of energy" [1].

Some viable sustainable energy sources include renewables such as wind and hydro powered turbines, to longer lasting nuclear and hydrogen sources. In comparison, in the UK, coal and oil have been steadily decreasing over time [2] - the last coal power plant was shut down on 30th September 2024 signalling a new era for the UK's future in energy.

The UK government also aims to increase spending in industries to increase exports and innovation efforts in hydrogen and its related technologies and "consistently feature in the top 10 countries globally for hydrogen technology patent rates" [1].

Although hydrogen as a renewable energy carrier is incredibly rewarding: "the chemical energy per mass of hydrogen (142 MJ kg^{-1}) is at least three times larger than that of other chemical fuels" [3] hydrogen fuel storage methods prove to be a major roadblock for using it as a widespread fuel within industries and personal use (for instance in cargo ships or in cars).

Most vehicles store hydrogen fuel cells in large high pressure gas cylinders. The industries target for these cylinders is 110 kg at 70 MPa (700 bar) [3]. The cylinders must be built to withstand the internal pressure as well as not be a major risk in the catastrophic event of a car crash or any other incident since hydrogen is a highly flammable gas.

This requirement can raise the weight of the overall fuel system and can act negatively to the overall efficiency of the vehicle in terms of top speed vs power.

When considering other forms of hydrogen storage, it can be stored in its liquid form. This requires extremely low temperatures of 21.2 K and within a closed system pressures can reach high levels (10,000 bar) [3].

Hydrogen can also be stored as a hydride - chemically bonded to metals in a reversible reaction. This reaction can cause strain within the material and the volume of hydrogen absorbed is set by its enthalpy change of both the hydrogen and metal [3]. Therefore, maintaining a fixed temperature within the system during fuelling is extremely important.

Another alternative is utilising metal organic frameworks (MOFs) to act as a sponge for hydrogen. MOFs are a nanoporous material consisting of metal ions and organic linking

branches that utilise the fact that hydrogen adsorbs on solid surfaces via van der Waals-type weak physisorption [3]. MOFs are able to store a higher volume of hydrogen compared to a solid block of material due to its porous nature; leading to its high surface area to volume ratio.

The properties of metal organic frameworks differ depending on the metal used and the structures in which the ions are held in. Additionally, they operate best as a hydrogen storage system in temperatures below 90 K and under high pressures (even at 500 bar it was observed full saturation was not achieved) [4].

Neural networks model the human brain's way of recognizing patterns through interconnected layers. They handle complex relationships well and are versatile, capable of tasks like predicting continuous values (regression) and categorizing data (classification). However, they need lots of data and computing power, and can overfit without proper regulation.

Decision trees split data based on feature values to make decisions, travelling along various nodes and decision functions in descending priority. They're easy to interpret and handle various data types, being useful in both regression and classification. However, decision trees and their variants like random forests and extremely randomised trees can be prone to overfitting.

Regression methods predict variables in a linear relationship – providing data that is easily interpretable since it can be plotted in a graph with clear spread of data points (and therefore error) against a line of best fit. This method can be altered for a more optimal output by using weights and biases.[5]

1.1. Aim and Objectives

1. Create an algorithm able to preprocess and featurise data into a format in which the neural network is easily able to process and utilise, to reduce the time spent manually processing multiple data sets.
2. Identify potential metal organic frameworks (MOFs) for hydrogen storage via the use of machine learning such as neural network to reduce the cost of the experimental process of manufacturing and testing MOFs and to aid in the innovation of hydrogen fuel.
3. Determine the optimal algorithm and set of properties in predicting potential MOFs for hydrogen storage to optimise the prediction process and therefore further reduce the cost of time and budget towards testing MOFs.
 - a. These properties are based on features such as topology, chemical features or electro-charged based properties.

2. LITERATURE REVIEW

The main paper studied in preparation was *Predicting hydrogen storage in MOFs via machine learning by Alauddin Ahmed and Donald J. Siegel* [6]. The paper covers different machine learning models used to predict the hydrogen storage capacity of various MOFs from extremely randomised trees, linear and ridge regressions, and other bagging algorithm variants.

The paper concludes that the extremely randomised trees (ERT) method is the most accurate method and also suits the context of the MOF structure extremely well since the models do not require large amounts of input data – suitable for theoretical MOF structures which have yet to undergo manufacturing. Boosted decision trees yield similar results as well as the random forest method. These methods are based on decision trees whilst linear and ridge regression methods had a poorer performance and support vector machines produced the worst results from the tested algorithms.

The accuracy and error levels of the ERT method produced R^2 values exceeding 0.997 for gravimetric capacities (the amount of H_2 adsorbed per unit mass) and error levels of 0.14 wt % - volumetric capacities had slightly higher errors, ranging from 1.1 g- H_2 L $^{-1}$ on average. In comparison, linear regression produced R^2 values ranging from 0.913 to 0.992 and error levels significantly higher than the tree-based methods.

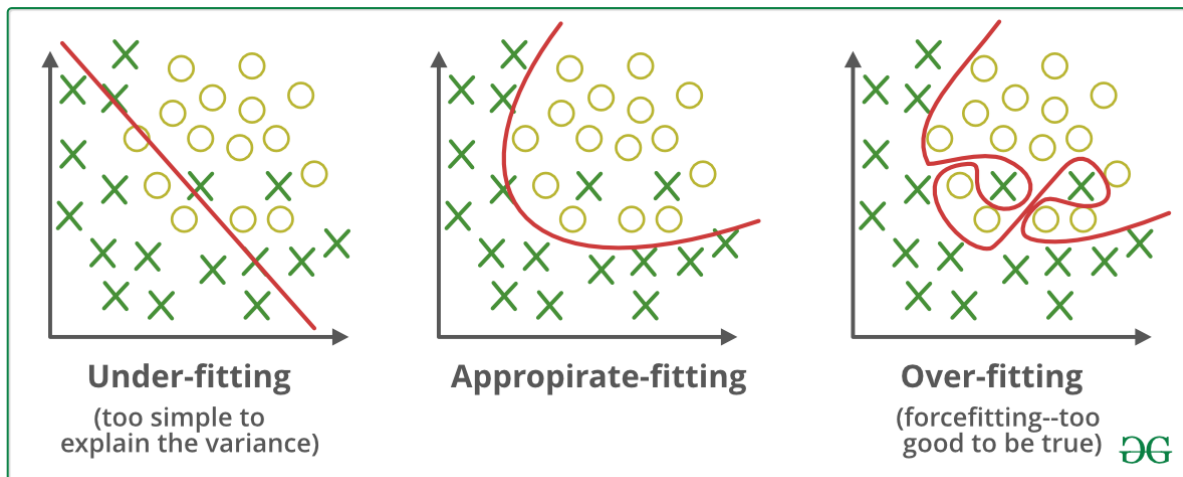


Figure 1: Underfitting and Overfitting [7]

Underfitting and overfitting are two aspects generally responsible for unsatisfactory results of the neural network. Underfitting is usually due to the lack of data used to train the model or the oversimplification of the algorithm. On the other hand, overfitting occurs when the model is too complex and there is high variance (sensitivity to changes) and low bias (predictions follow the provided data closely). Underfitting can be avoided by providing more data although care must be taken to ensure that the data has been scaled and is appropriate for the algorithm – it has been featurised. Overfitting can be reduced by allowing the network to

focus on more meaningful patterns in the overall data rather than placing a large bias on small changes within the training data. Reducing the complexity of the system is also a good method however it should be noted that in this project, underfitting is of a higher likelihood of occurring than overfitting. [8]

Other limitations on the neural network originate from the Chahine rule which suggests that increasing the specific surface area of a nanoporous material enhances its gravimetric hydrogen uptake. [9] The Chahine rule generally overpredicts hydrogen capacities for high-surface-area MOFs and so identifying other key features that influence hydrogen uptake such as the availability of electronegativity sites and other open metal sites of MOFs formed due to the topology and chemical features of the structure.

The paper also highlights the fact that synthesising high-capacity MOFs is quite difficult and sometimes not feasible to create in practice. In addition, even if the predicted hydrogen storage capacity for a certain MOF is high, during activation, the MOF may experience framework collapse which will reduce its available space for hydrogen.[6]

3. PROGRESS UPDATE

3.1. Methodology

For the programming of the algorithms and other software-based solutions: agile, iterative and V-shaped models of software development methods were considered.

The agile model cycles through stages of development and testing, taking in customer feedback and altering the requirements and plans for the programming stages in small incremental changes.

The iterative model is similar to the agile model, building upon the initially created code with each new piece of feedback. Its response to issues raised during implementation and testing helps fix any errors quickly.

The V-shaped model is more structured compared to the two other methods in which each phase must be completed before moving to the next stage. Each testing stage directly corresponds to a development stage allowing for any feedback to be implemented into the program but is less flexible overall compared to the agile and iterative models.

Out of the three, the iterative approach was chosen since it provides a steady structure as the V-shaped method would but still holds the flexibility and quick responses to any changes to the requirements or algorithm structures.

3.2. Featurisation

The featurisation code was written in python and heavily utilises the csv python library. The code mainly focuses on cleaning the data and removing any unnecessary data points such as duplicates with as little user input as possible in order to increase the overall processing speed.

The data fed into the featurisation code and later will be used as training data for the neural networks is that used in the paper *Predicting hydrogen storage in MOFs via machine learning* by Alauddin Ahmed and Donald J. Siegel [6].

The data for the MOFs have been provided for two different environments: a pressure swing (PS); and a temperature-pressure swing (TPS).

The pressure swing data focuses on an environment that is isothermal: $T = 77\text{ K}$, in which the pressure has a swing range of 100 and 5 bar.

The temperature-pressure swing data considers the conditions where the pressure and temperature holds a variation of a full condition: $P = 100\text{ bar}$ and $T = 77\text{ K}$, to its empty condition in which: $P = 5\text{ bar}$ and $T = 160\text{ K}$.

At present, the project will only consider and work with the isothermal PS data since it has been deemed easier to start with a more simple machine learning model with less input data.

As the project progresses, more complex algorithms can be considered and so the TPS data with more factors to take into account can be used.

This data used as training data also provides properties such as the gravimetric (mass of the stored H₂ per combined mass of the H₂ and MOF) and volumetric capacity (grams of H₂ per liter of MOF) of the MOFs. This combined with other features such as the MOF surface area, pore properties and densities – among others – will allow the featurisation code to calculate other attributes like the fraction of volume in the MOF accessible for hydrogen storage via library packages like zeo++ [10].

Summarily, the featurisation data will process the data from the paper *Predicting hydrogen storage in MOFs via machine learning by Alauddin Ahmed and Donald J. Siegel* [6]. It will remove any duplicate data and calculate other features such as accessible volume and structure data for future analysis of its effects against hydrogen intake (gravimetric and volumetric capacity) using the zeo++ library.

3.3. Program Planning

Currently, aspects of the machine learning model have been planned out. The method chosen for the first algorithm is linear regression. The python library scikit-learn will be predominately used for the base algorithms for linear regression and the simple decision tree method.

Linear regression is based over the mathematical equation:

$$h(x) = \theta_0 + \theta_1 x_1 + \dots + \theta_n x_n \quad \text{Equation 1}$$

Where $h(x)$ is the predicted value, θ_0 is the bias term, $\theta_1, \dots, \theta_n$ are the model parameters and $x_1, x_2, \dots, x_n$ are the feature values.

These predicted values can be plotted onto a graph with the variable property against the gravimetric capacity (chosen instead of volumetric since volumetric capacity assumes the MOF is a monolith).

A regression line (line of best fit) can then be calculated by finding the points in which the error between the training data (observed data) and predicted data is the lowest. This can also be done using this equation for the cost function:

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h(x^i) - y^i)^2 \quad \text{Equation 2}$$

Where the cost function $J(\theta)$ is at its lowest.

In Equation 2, m represents the total number of training examples, x and y are our feature values and $h(x)$ refers back to Equation 1.

For future progress in which the algorithms will be more finely adjusted for optimum results, the code is expected to get more complicated but the base code currently planned will be the foundation upon which algorithms like random forests and ridge regression will be developed.

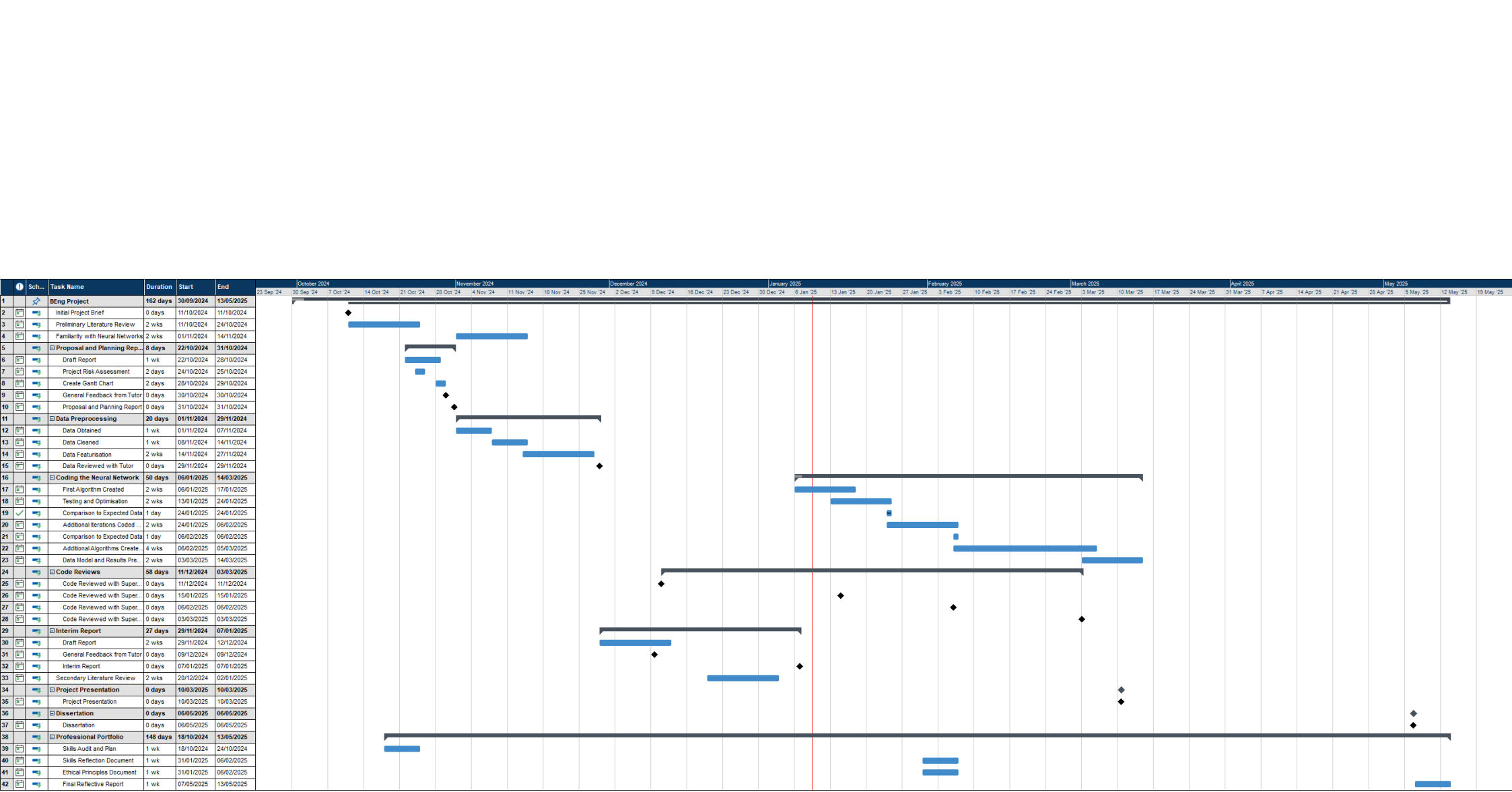
4. PLANNING UPDATE

Presently, the project is 2 weeks behind the initial schedule. Featurisation of data has been completed and the neural network code is within the planning stage. It should also be noted that code used to featurise data will be updated and altered in parallel development to the neural network code depending on the data needs of the algorithm: e.g. data items scaled to appropriate units, open metal sides are to be calculated via `zeo++` libraries etc.

One particular set back highlighted was the inaccessibility of key data needed during a meeting with the supervisor. The data was held online and was not back-up to a local hard drive and so was unattainable without an internet connection. To prevent this in the future, the frequency of offline backups was increased and a remote GitHub repository was created (which was also linked to an online connection).

5. CONCLUSIONS

Overall, the project focuses on building upon well explored models and predictions to aid the development of hydrogen storage as a renewable energy carrier. Expected limitations and roadblocks in the neural network training are to be underfitting which can be solved by improving the coded algorithm or introducing more variables and data such as the temperature-pressure swing data.



6. REFERENCES

- [1] (2023). *Powering Up Britain: Net Zero Growth Plan*.
- [2] (2023). *UK ENERGY IN BRIEF 2023*.
- [3] A. Züttel and L. Schlapbach, "Hydrogen-storage materials for mobile applications," *Nature (London)*, vol. 414, no. 6861, pp. 353-358, 2001, doi: 10.1038/35104634.
- [4] L. Zhang *et al.*, "Fundamentals of hydrogen storage in nanoporous materials," *Progress in energy*, vol. 4, no. 4, p. 42013, 2022, doi: 10.1088/2516-1083/ac8d44.
- [5] S. Henrard, N. Speybroeck, and C. Hermans, "Classification and regression tree analysis vs. multivariable linear and logistic regression methods as statistical tools for studying haemophilia," *Haemophilia : the official journal of the World Federation of Hemophilia*, vol. 21, no. 6, pp. 715-722, 2015, doi: 10.1111/hae.12778.
- [6] A. Ahmed and D. J. Siegel, "Predicting hydrogen storage in MOFs via machine learning," *Patterns (New York, N.Y.)*, vol. 2, no. 7, pp. 100291-100291, 2021, doi: 10.1016/j.patter.2021.100291.
- [7] G. f. Geeks, <https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/>. Underfitting and Overfitting. Available: <https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/>.
- [8] G. f. Geeks. "ML | Underfitting and Overfitting." <https://www.geeksforgeeks.org/underfitting-and-overfitting-in-machine-learning/> (accessed 2024).
- [9] D. P. Broom, C. J. Webb, G. S. Fanourgakis, G. E. Froudakis, P. N. Trikalitis, and M. Hirscher, "Concepts for improving hydrogen storage in nanoporous materials," *International journal of hydrogen energy*, vol. 44, no. 15, pp. 7768-7779, 2019, doi: 10.1016/j.ijhydene.2019.01.224.
- [10] M. Haranczyk and K. Jariwala. "zeo++ Examples." <https://www.maciejharanczyk.info/Zeo++examples.html> (accessed).